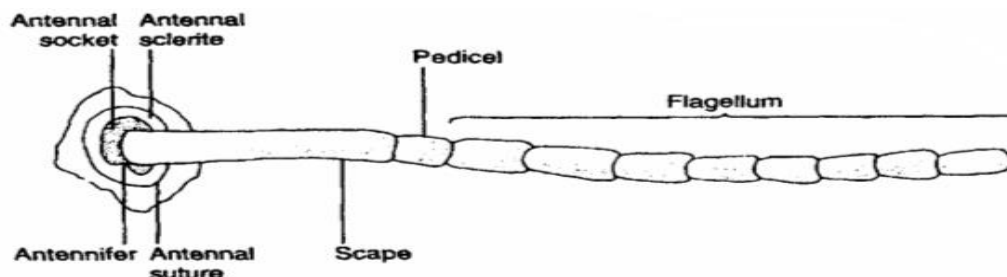


❖ Antennae having three segments:

1. Scape
2. Pedicel
3. Flagellum



- **Scape:** it is the first segment, larger than the other segments and is the basal stalk.
- **Pedicel:** It is the second segment, nearly always contains a sensory organ known as Johnston's organ, which responds to movement of the distal part of the antenna related to the pedicel.
- **Flagellum:** It is the filamentous and multisegmented (with many flagellomeres) but may be reduced or variously modified.

❖ TYPES OF ANTENNAE:

Type	Remarks	Example
Aristate	Pouch-like	House fly
Setaceous	Bristle-like	Dragonfly/damselfly
Filiform	Thread-like	Grasshopper
Moniliform	Bead-like	Termites
Serrate	Saw-toothed	Beetles
Pectinate	Comb-like	Moths, sawflies, beetles
Capitate	Head like (less enlarged at the tip would be clavate -clublike)	Butterflies
Geniculate	Elbowed	Honeybees
Lamellate	Plate-like	Beetles
Plumose	Plumed or feather-like	Mosquito

❖ DIFFERENT KIND OF ORGANS:

1. **Organs of smell:** In some insects the smell organs (sensoria) are situated in the antennae by which they recognize their food etc. **e.g., Ants, honeybee.**

2. Organs of taste: Some insects bear taste hair on their antennae by which they know the taste of their food e.g., cockroach.

3. Stridulatorial organs: Sound producing organs are in the antennae of some insects belonging to the orders-coleopteran and orthopteran e.g., cricket.

4. Organs of chordotonal: The hearing organs often known as Johnston's organs are situated in the second segment (pedicel) of antenna e.g., male mosquito.

5. Sexual characters: In some of the insects belonging to the order Diptera and Hemiptera, the antennae are found of different type in male and female viz. mosquito.

6. Other functions: In some larvae the antennae are adapted for seizing the prey e.g., *Chaborus*, while in other insects they are used for holding the females. e.g., the male of *meloe*. The butterflies are having some transmitting and receiving organs in their antennae.

❖ Mouth parts:

- **Important** for insect identification.
- Provides information on feeding habits and types of damage

Mouth Part	Remarks
Labrum or upper lip	It is with a ventral surface called the epipharynx
Hypopharynx	A tongue like structure
Mandibles or jaws	Crush food and may be used for defense
Maxillae (Maxilla)	Hold the food
Labium or lower lip	

❖ Types:

- **Chewing and biting parts (Mandibulate):** Used to chew holes in leaves, bore in stems. These types of mouth parts are present in cockroach, grasshopper, cricket, beetles and earwigs.
- **Piercing and sucking type:**
 - ✓ Examples: **mosquitoes, stink bugs, etc.**
 - ✓ True bugs (Hemiptera), thrips (Thysanoptera), fleas (Siphonaptera) and sucking lice (Psocodea: Anoplura).

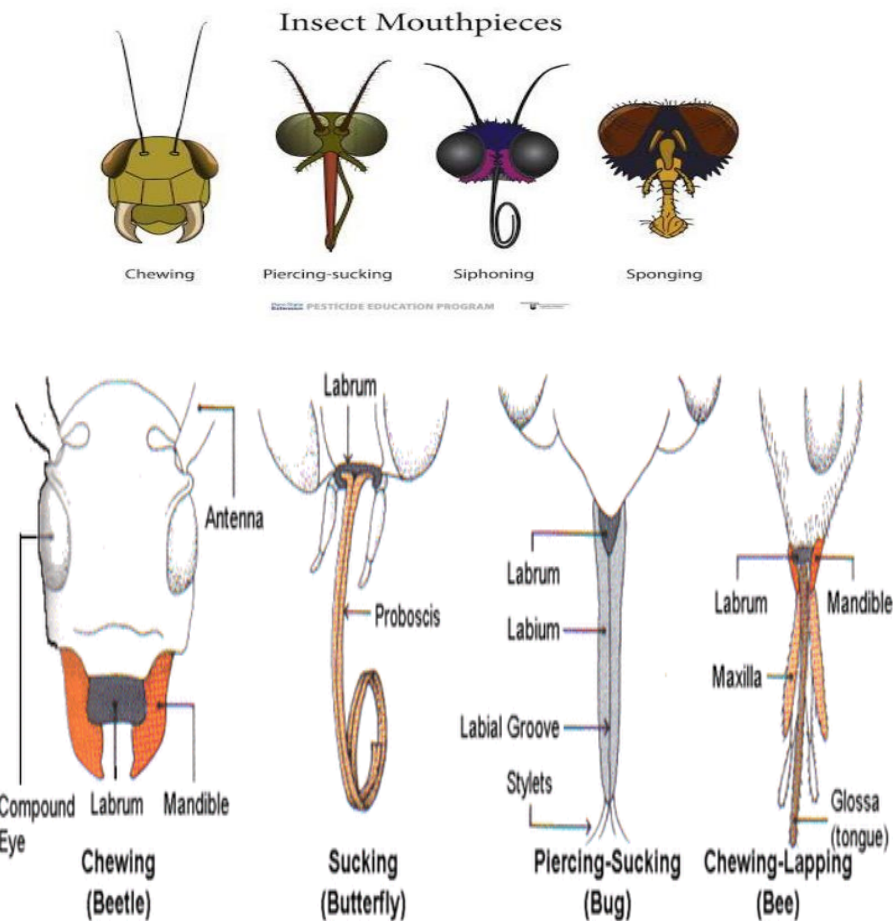
- ✓ In each order different mouth part components form needle like stylets capable of piercing the plant or animal tissues upon which the insect feeds.
- ✓ Bugs have extremely long thin paired mandibular and maxillary stylets, which fit together to form a flexible stylet bundle containing a food canal and a salivary canal.
- **Chewing and piercing-sucking:** Rasp (scrap) surfaces of leave, suck up sap. Example: thrips, Thrips have 3 stylets- paired maxillary stylets (Laciniae) plus the left mandibular one.
- **Sponging Type:**
 - ✓ Modified for liquids or solid foods.
 - ✓ Solid foods must be dissolved by salivary secretions.
 - ✓ **Example: House fly**
- **Siphoning Type:**
 - ✓ Mouthparts form a sucking tube (proboscis).
 - ✓ Modified for uptake of nectar/liquids.
 - ✓ Coiled beneath head when not in use.
 - ✓ **Examples: Butterflies and moths**
- **Rasping-Lapping Type:**
 - ✓ Modified to use liquid or semi-liquid foods.
 - ✓ Some mouthpart components function for chewing
- **Mold wax**
- **Grasping prey**
- **Cutting flowers**
 - ✓ Other components form the proboscis 'lapping' surface.
 - ✓ **Lapping** is a mode of feeding in which liquid or semi-liquid food adhering to a protrusible organ or tongue is transferred from substrate to mouth.

❖ TYPES OF MOUTH PARTS:

Types of Mouth Parts

Examples

Biting and Chewing	Locust, Grasshopper, Cricket
Piercing & sucking type	Mosquito, Aphid, Bugs, Leaf hopper
Sponging Type	Housefly
Siphoning type	Butterfly
Rasping and sucking	Thrips
Rasping and lapping	Honeybee



❖ **Thorax:**

- **Divided** into 3 regions.

1. **Prothorax:** Bears 1 pair of legs
2. **Mesothorax:** Bears 1 pair of legs, 1 pair of wings
3. **Metathorax:** Bears 1 pair of legs, 1 pair of wings

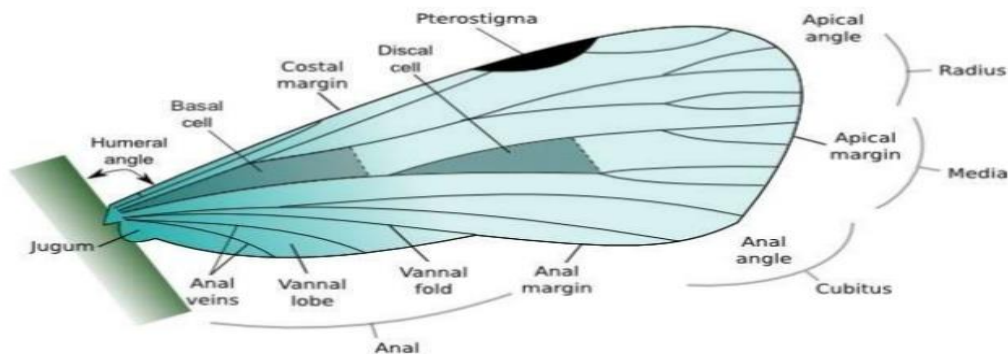
- ❖ **Wings:** Number of wings varies by species. In general, have 2 pairs.

- **Mesothoracic wing:** Forewing
- **Metathoracic wing:** Hindwing

❖ Functions

- Locomotion.
- Protection.
- Camouflage

❖ Wing modifications:



❖ TYPE OF WINGS:

Types	Remarks	Examples
Halters (Halter)	Knob-like reduced hind wings of Diptera	House fly
Elytra (Elytron)	Hardened, protective forewings of Coleoptera	Beetle and weevils
Hemelytra	Half-hardened, half-membranous forewings of Hemiptera	Red cotton bug
Membranous wings		Dragon flies, bees, wasp, grasshopper
Fringed wings	Modified wing structure of the Thysanoptera	Thrips
Scales and hairs		Lepidoptera, Trichoptera, some Diptera Moth and butterflies
Tegmina		Cockroach